

E7 Introduction to Computer Programming for Scientists and Engineers

Discussion week 12

Solving systems of nonlinear equations with FPI

Example: Find a solution to the following system of equations.

$$\begin{cases} 10\xi_1 + \xi_2 = 1 - \mathbf{\sin}(\xi_1 + \xi_2) \\ -\xi_1 + 10\xi_2 = 2 + \mathbf{\cos}(\xi_1 - \xi_2) \end{cases}$$

Candidate fixed point function #1.

$$\begin{cases} \xi_1 = 10\xi_2 - 2 - \mathbf{cos}(\xi_1 - \xi_2) \\ \xi_2 = -10\xi_1 + 1 - \mathbf{sin}(\xi_1 + \xi_2) \end{cases}$$

Candidate fixed point function #2. g_A

$$\begin{cases} \xi_1 = 0.1(1 - \xi_2 - \mathbf{sin}(\xi_1 + \xi_2)) \\ \xi_2 = 0.1(2 + \xi_1 + \mathbf{cos}(\xi_1 - \xi_2)) \end{cases}$$

Candidate fixed point function #3. g_B

- Change of variables.

$$\begin{array}{lcl} \eta_1 = \xi_1 + \xi_2 & & \xi_1 = (\eta_1 + \eta_2)/2 \\ \eta_2 = \xi_1 - \xi_2 & \Longleftrightarrow & \xi_2 = (\eta_1 - \eta_2)/2 \end{array}$$

- g_B

$$\begin{cases} \eta_1 = (1/11) \left(-9\eta_2 + 2 - 2 \sin(\eta_1) \right) \\ \eta_2 = (1/11) \left(9\eta_1 - 4 - 2 \cos(\eta_2) \right) \end{cases}$$